

Relational Algebra

txt2tex example

June 13, 2026

[*Title, Author, MemberName, Genre*]

Book

bookId : $\mathbb{N}$
title : *Title*
author : *Author*
genre : *Genre*
pages : $\mathbb{N}$
year : $\mathbb{N}$

Member

memberId : $\mathbb{N}$
name : *MemberName*

Loan

loanId : $\mathbb{N}$
bookId : $\mathbb{N}$
memberId : $\mathbb{N}$
dueYear : $\mathbb{N}$

Restriction (σ)

$\text{Restrict}_{\text{pages} \geq 500}(\text{Book})$
 $\text{Restrict}_{\text{year} \geq 1990}(\text{Restrict}_{\text{pages} \geq 500}(\text{Book}))$

Projection (π)

$\text{Project}_{\text{title}, \text{author}}(\text{Book})$
 $\text{Project}_{\text{title}, \text{author}}(\text{Restrict}_{\text{pages} \geq 500}(\text{Book}))$

Renaming (ρ [new/old])

$\text{Project}_{\text{id}, \text{title}, \text{author}, \text{genre}, \text{pages}, \text{year}}(\text{Book}[\text{id}/\text{bookId}])$
 $\text{Project}_{\text{id}, \text{writer}, \text{genre}, \text{pages}, \text{year}}(\text{Book}[\text{id}/\text{bookId}, \text{writer}/\text{author}])$

Natural Join and Theta-Join ($\Join$)

$\text{Join}(\text{Loan}, \text{Book})$
 $\text{Project}_{\text{id}, \text{memberId}, \text{dueYear}, \text{title}, \text{author}, \text{genre}, \text{pages}, \text{year}}(\text{Join}(\text{Loan}[\text{id}/\text{bookId}], \text{Book}[\text{id}/\text{bookId}]))$
 $\text{Join}_{\text{Member.memberId} = \text{Loan.memberId}}(\text{Member}, \text{Loan})$

Division (div)

Loan div *Book*

Combined Query

Project_{*title,author*}(Restrict_{*pages* ≥ 500}(*Book*))